CORE – 7: Discrete Mathematical Structures – Unit- 2

Basic Principles of Counting:

Sum Rule Principle: Assume some event E can occur in m ways and a second event F can occur in n ways, and suppose both events cannot occur simultaneously. Then E or F can occur in m + n ways.

In general, if there are n events and no two events occurs in same time then the event can occur in n_1+n_2.....n ways.

Example: If 8 male professor and 5 female professor teaching DMS then the student can choose professor in 8+5=13 ways.

Product Rule Principle: Suppose there is an event E which can occur in m ways and, independent of this event, there is a second event F which can occur in n ways. Then combinations of E and F can occur in mn ways.

In general, if there are n events occurring independently then all events can occur in the order indicated as $n_1 \times n_2 \times n_3$.....n ways.

Example: In class, there are 4 boys and 10 girls if a boy and a girl have to be chosen for the class monitor, the students can choose class monitor in 4 x 10 = 40 ways.

Permutation: Any arrangement of a set of n objects in a given order is called Permutation of Object. Any arrangement of any r ≤ n of these objects in a given order is called an r-permutation or a permutation of n object taken r at a time. It is enoted by ${}^nP_r$ or P(n,r).

$$P(n,r) = \frac{n!}{(n-r)!}$$

Theorem: Prove that the number of permutations of n things taken all at a time is n!.
Proof: We know that

$$P(n,n) = \frac{n!}{(n-n)!} = \frac{n!}{0!} = \frac{n!}{1} = n!$$

Permutation with Restrictions: The number of permutations of n different objects taken r at a time in which q particular objects do not occur is P(n-q,r).

The number of permutations of n different objects taken r at a time in which q particular objects are present is P(n-q,r-q) x q!.

Example: How many 6-digit numbers can be formed by using the digits 0, 1, 2, 3, 4, 5, 6, 7, 8 if every number is to start with '30' with no digit repeated?

Solution: All the numbers begin with '30.'So, we have to choose 4-digits from the remaining 7-digits.

∴ Total number of numbers that begins with '30' is P(7,4) = 840.

$$\Rightarrow C(n,r) = \frac{P(n,r)}{r!}$$

$$\Rightarrow C(n,r) = \frac{n!}{r!(n-r)!}$$ Hence, proved.

Example: A farmer purchased 3 cows, 2 pigs, and 4 hens from a man who has 6 cows, 5 pigs, and 8 hens. Find the number of choices that the farmer has.

The farmer can choose the cows in C (6, 3) ways, the pigs in C (5, 2) ways, and the hens in C (8, 4) ways. Thus the number of choices that the farmer has is equals to

$$C(6,3) \times C(5,2) \times C(8,4) = \frac{\cancel{6}.5.4 \,.\, 5.4 \,.\, \cancel{8}.7.\cancel{6}.5}{\cancel{3.2}.1 \,.\, \cancel{2.1 \,.\, 4.3.2}.1} = \frac{5.4.5.4.7.5}{1} = 14000$$

The Pigeonhole Principle: If n pigeonholes are occupied by n+1 or more pigeons, then at least one pigeonhole is occupied by more than one pigeon.

Generalized pigeonhole principle: - If n pigeonholes are occupied by kn+1 or more pigeons, where k is a positive integer, then at least one pigeonhole is occupied by k+1 or more pigeons.

Example1: Find the minimum number of students in a class to be sure that three of them are born in the same month.

Solution: Here n = 12 months are the Pigeonholes and k + 1 = 3

$\Rightarrow$ K = 2

$\Rightarrow$ Kn + 1 = 2 x 12 + 1 = 25

So, the minimum number of students in a class to be sure that three of them are born in the same month is 25.

Recurrence Relations: A recurrence relation is an equation that recursively defines a sequence where the next term is a function of the previous terms (Expressing Fn as some combination of Fi with i<n).

Example: Fibonacci series, $F_n = F_{n-1} + F_{n-2}$.

A recurrence relation is also called a difference equation, and these two terms are used interchangeably.

Order of the Recurrence Relation: The order of the recurrence relation or difference equation is defined to be the **difference between the highest and lowest subscripts** of f(x).

Example1: The equation $13a_r + 20a_{r-1} = 0$ is a first order recurrence relation.

Example2: $F_n = F_{n-1} + F_{n-2.}$ Order of this recurrence relation is n – (n-2) = 2

Degree of the Difference Equation: The degree of a difference equation is defined to be the highest power of f (x).

Example1: The equation $y^3_{k+3} + 2y^2_{k+2} + 2y_{k+1} = 0$ has the degree 3, as the highest power of y_k is 3. Its order is (k+3) – (k+1) = 2.

Linear Recurrence Relations with Constant Coefficients: A Recurrence Relations is called linear if its degree is one.

The general form of linear recurrence relation with constant coefficient is

$a_n = C_1 a_{n-1} + C_2 a_{n-2} + \cdots + C_k a_{n-k} + f(n)$ ---------- eqn(0)

Where $C_1, C_2 \ldots\ldots C_k$ are real numbers and $C_k \neq 0$.

If f(n) = 0, then eqn(0) is homogeneous otherwise non-homogeneous.

Example1: The equation $a_{n+3} + 6a_{n+2} + 12a_{n+1} + 8a_n = 0$ is a homogeneous linear equation of order 3.

Example2: The equation $a_{n+2} - 4a_{n+1} + 4a_n = 3n + 2^n$ is a non-homogeneous linear equation of order 2.

Solving Linear Homogeneous Recurrence Relations with Constant Coefficients:

Consider the equation of the form

$a_n = C_1 a_{n-1} + C_2 a_{n-2} + \cdots + C_k a_{n-k}$ ---eqn(1), Putting $a_n = r^n$ where r is a constant, we get

⇨ $r^n = C_1 r^{n-1} + C_2 r^{n-2} + \cdots + C_k r^{n-k}$ Dividing by r^{n-k}, we get

⇨ $r^k = C_1 r^{k-1} + C_2 r^{k-2} + \cdots + C_k$

⇨ $r^k - C_1 r^{k-1} - C_2 r^{k-2} - \cdots - C_k = 0$ -----eqn(2) **(Characteristic equation)**

Consequently, the sequence $\{a_n\}$ with $a_n = r^n$ is a solution of eqn(1) if and only if r is a solution of eqn(2).

eqn(2) is known as the Characteristic equation and the values of r after finding the solution are known as the characteristic roots.

Example1: Solve the difference equation $a_n = 6a_{n-1} - 8a_{n-2}$

Solution: Given $a_n = 6a_{n-1} + 8a_{n-2}$ Putting $a_n = r^n$ where r is a constant, we get

$r^n - 6r^{n-1} - 8r^{n-2} = 0$ Dividing by r^{n-2}, we get

⇨ $r^2 - 6r - 8 = 0$

⇨ $r^2 - 4r - 2r - 8 = 0$

⇨ $r(r - 4) - 2(r + 4) = 0$

⇨ $(r - 4)(r - 2) = 0$

⇨ $r = 4$ or $r = 2$

Therefore, $a_n = C_1 2^n + C_2 4^n$

Example2: Solve the difference equation $a_n = 6a_{n-1} + 8a_{n-2}$, given $a_0 = 0$ and $a_1 = 4$.

Solution: Given $a_n = 6a_{n-1} + 8a_{n-2}$ Putting $a_n = r^n$ where r is a constant

We get, $r^n - 6r^{n-1} - 8r^{n-2} = 0$ Dividing by r^{n-2}

⇨ $r^2 - 6r - 8 = 0$

⇨ $r^2 – 4r – 2r – 8 = 0$

⇨ $r(r – 4) – 2(r + 4) = 0$

⇨ $(r – 4)(r – 2) = 0$

⇨ $r = 4$ or $r = 2$

Therefore, $a_n = C_1 2^n + C_2 4^n$ ---------- eqn(3)

⇨ $a_0 = C_1 2^0 + C_2 4^0$ (putting n = 0)

⇨ $0 = C_1 + C_2$

⇨ $C_1 = –C_2$ ----------eqn(4)

Now, $a_n = C_1 2^n + C_2 4^n$ putting n = 1

⇨ $a_1 = C_1 2^1 + C_2 4^1$

⇨ $4 = 2C_1 + 4C_2$

⇨ $4 = –2C_2 + 4C_2$ (Putting $C_1 = –C_2$)

⇨ $4 = 2C_2$

⇨ $C_2 = 2$ ---------- eqn(5)

⇨ $C_1 = –2$ ---------- eqn(6)

Putting the value of C_1 and C_2 in eqn(3) we get the solution

$a_n = –2x2^n + 2x4^n$ or

$a_n = –2^{n+1} + 2x4^n$ (Solution)

Solving Non-homogeneous Linear Recurrence Relations with Constant Coefficients:

A recurrence relation is called non-homogeneous if it is in the form

$a_n = C_1 a_{n-1} + C_2 a_{n-2} + \cdots + C_k a_{n-k} + f(n)$ where $f(n) \neq 0$

Its associated homogeneous recurrence relation is $a_n = C_1 a_{n-1} + C_2 a_{n-2} + \cdots + C_k a_{n-k}$

The solution (a_n) of a non-homogeneous recurrence relation has two parts. First part is the solution (a_h) of the associated homogeneous recurrence relation and the second part is the particular solution (a_p).

$$a_n = a_h + a_p$$

Solution to the first part is done using the procedures discussed in the previous section. To find the particular solution, we find an appropriate trial solution. Trial solution for some functions f(n) is shown in the table below.

f(n)	Trial solutions	
C	A	A, A_2, A_1, A_0, B, and C are constants
n	$A_1n + A_0$	
n^2	$A_2n^2 + A_1n + A_0$	
C^n	AC^n	
BC^n	AnC^n	

Example: Solve the recurrence relation

$a_{n+2} + 3a_{n+1} + 2a_n = 3^n$ for n >= 0 where $a_0 = 0$ and $a_1 = 1$

Solution: This is a linear non-homogeneous relation, where the associated homogeneous equation is

$a_{n+2} + 3a_{n+1} + 2a_n = 0$ and $f(n) = 3^n$

Put $a_n = r^n$

So, $r^{n+2} + 3\,r^{n+1} + 2\,r^n = 0$ Deviding by r^n, we get

⇨ $r^2 + 3r + 2 = 0$

⇨ $r^2 + 2r + r + 2 = 0$

⇨ $r(r+2) + 1(r+2) = 0$

⇨ $(r+2)(r+1) = 0$

⇨ $r = -2$ or $r = -1$

⇨ $a_h = A.(-2)^n + B.(-1)^n$ ----------eqn(1) where A and B are constants.

Since $f(n)=3^n$, i.e. of the form C^n, a reasonable trial solution of a_t will be AC^n

$$a_p = AC^n = A3^n$$

The given recurrence relation is $a_{n+2} + 3a_{n+1} + 2a_n = 3^n$

So, by substituting $A3^n$ to each position of the given recurrence relation, we get

$A3^{n+2} + 3A3^{n+1} + 2A3^n = 3^n$ dividing each side by 3^n we get,

$A3^2 + 3A3^1 + 2A = 1$

⇨ $9A + 9A + 2A = 1$

⇨ $20A = 1$

⇨ $A = 1/20$

Guess value of $a_p = A3^n$

⇨ $a_p = 1/20.3^n$ ----------eqn(2)

Hence, Total solution $a_n = a_h + a_p$ putting value of eqn(1) and eqn(2), we get

⇨ $a_n = A.(-2)^n + B.(-1)^n + 1/20.3^n$ ----------eqn(3), Putting $a_0 = 0$, we get

⇨ $a_0 = A.(-2)^0 + B.(-1)^0 + 1/20.3^0$

⇨ $0 = A + B + 1/20$

⇨ $A + B = -1/20$ ----------eqn(4), Putting $a_1 = 1$ in eqn(3), we get

$a_1 = A.(-2)^1 + B.(-1)^1 + 1/20.3^1$

⇨ $1 = -2A – B + 3/20$

⇨ -2A - B = 17/20 ----------eqn(5)

Adding eqn(4) and eqn(5) we get

-A = 16/20

⇨ A = -4/5 ----------eqn(6)

Putting the value of A in eqn(4) we get

-4/5 + B = -1/20

⇨ B = 4/5 - 1/20

⇨ B = 3/4 ----------eqn(7)

The total solution which is represented by eqn(3) is

$a_n = A.(-2)^n + B.(-1)^n + 1/20.3^n$ Now putting the value of A and B here we get

$a_n = -4/5.(-2)^n + 3/4.(-1)^n + 1/20.3^n$ (Total solution)

Solving Recurrence Relation using Generating Functions:

The generating function for the sequence $a_0, a_1, a_2, \ldots, a_k, \ldots$ of real numbers is the infinite series

$$G(x) = a_0 + a_1 x + \cdots + a_k x^k + \cdots = \sum_{k=0}^{\infty} a_k x^k$$

The following table shows some generic functions and their equivalent numeric functions.

Generating function	Numeric function
$G(x) = \frac{1}{1 - ax}$	a^r
$G(x) = \frac{1}{(1 - x)^2}$	r + 1
$G(x) = \frac{1}{(1 - ax)^2}$	$(r + 1)a^2$
$G(x) = \frac{x}{(1 - x)^2}$	r

Example: Solve the following recurrence relation using generating function.

$a_n - 2a_{n-1} - 3a_{n-2} = 0$ for n >= 2 where $a_0 = 3$ and $a_1 = 1$

Solution: Given $a_n - 2a_{n-1} - 3a_{n-2} = 0$ for n >= 2

Multiplying both sides by r^n we get

$a_n x^n - 2a_{n-1} x^n - 3a_{n-2} x^n = 0$

Taking the summation we get

$$\sum_{n=2}^{\infty} a_n x^n - 2\sum_{n=2}^{\infty} a_{n-1} x^n - 3\sum_{n=2}^{\infty} a_{n-2} x^n = 0 \quad \text{---eq}^n(1)$$

Now, the 1st term is:

$$\sum_{n=2}^{\infty} a_n x^n = a_2x^2 + a_3x^3 + \cdots\cdots\cdots$$

$$= a_0 + a_1x + a_2x^2 + a_3x^3 + \ldots\ldots\ldots\ldots\ldots - a_0 - a_1x$$

$$= a(x) - a_0 - a_1x \text{ ---------- eqn(2)}$$

2nd term:

$$\sum_{n=2}^{\infty} a_{n-1} x^n = a_1x^2 + a_2x^3 + \cdots\cdots\cdots$$

$$= a_0x + a_1x^2 + a_2x^3 + \ldots\ldots\ldots\ldots\ldots a_0x$$

$$= (a_0 + a_1x + a_2x^2 + \ldots\ldots\ldots\ldots\ldots)x - a_0x$$

$$= [a(x) - a_0]x \text{ ---------- eqn(3)}$$

3rd term:

$$\sum_{n=2}^{\infty} a_{n-2} x^n = a_0x^2 + a_1x^3 + a_2x^4 + \cdots\cdots\cdots$$

$$= (a_0 + a_1x + a_2x^2 + \ldots\ldots\ldots\ldots\ldots)x^2$$

$$= a(x)x^2 \text{ ---------- eqn(4)}$$

Putting the value of eqn(2), eqn(3) and eqn(4) in eqn(1), we get

$$a(x) - a_0 - a_1x - 2[a(x) - a_0]x - 3a(x)x^2 = 0$$

$$\Rightarrow a(x) - a_0 - a_1x - 2a(x).x + 2a_0x - 3a(x)x^2 = 0$$

Substituting the value of $a_0 = 3$ and $a_1 = 1$, we get

$$a(x) - 3 - x - 2a(x).x + 6x - 3a(x)x^2 = 0$$

$$\Rightarrow a(x)\,[1 - 2x - 3x^2] = 3 - 5x$$

$$\Rightarrow a(x) = \frac{3 - 5x}{1 - 2x - 3x^2} = \frac{3 - 5x}{1 - 3x + x - 3x^2}$$

$$= \frac{3 - 5x}{1 - 3x + x(1 - 3x)} = \frac{3 - 5x}{(1 - 3x)(1 + x)}$$

$$= \frac{1}{(1-3x)} + \frac{2}{(1+x)} \quad (\textbf{By partial fraction})$$

$\Rightarrow a^n = 3^n + 2(-1)^n$ (Ans)

Note: Taking 1–3x = 0 or x = 1/3, the expression $\frac{3-5x}{(1+x)}$ becomes 1

Taking 1 + x = 0 or x = –1, the expression $\frac{3-5x}{(1-3x)}$ becomes 2